

B. Tech. Project Mid-Semester Evaluation

Explainable Deep Learning Framework for Automated ECG Arrhythmia Detection and Classification

Presented by

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Introduction

Electrocardiogram (ECG)

Skin electrodes record the heart's electrical activity. Each beat shows a P wave, QRS complex and T wave; arrhythmias alter their morphology (shape) or rhythm (timing).

Why automate?

A 24 h Holter recording at ~70 bpm holds $\approx$ 100,000 beats. Manual beat-by-beat review is slow and error-prone, so automatic beat classification is needed.

What the framework must deliver

Accurate classification

per-class F1 on N, S, V, F

Explanation

sample-level heatmap per beat

Robustness

stable under real ECG noise

Deep models reach high accuracy but act as black boxes. This project pairs a CNN / CNN-LSTM classifier with Grad-CAM and SHAP, and quantitatively checks whether the explanations point at the clinically relevant waves.

ECG Beats & Arrhythmia Classes

P, QRS and T waves

P = atrial depolarisation; QRS = ventricular depolarisation (normal width < 120 ms); T = ventricular repolarisation. Normal PR interval 120–200 ms.

RR interval

Time between successive R peaks; heart rate = $60 / \text{RR}$ (s). A premature beat has a short previous RR and a long next RR (compensatory pause).

N – Normal {N, L, R, e, j}

Normal, bundle-branch-block and escape beats. $\approx 90\%$ of all beats, so the data are heavily imbalanced.

S – Supraventricular {A, a, J, S}

Premature beats from the atria / AV junction. QRS is near-normal, so detection relies on the RR timing and an abnormal or hidden P wave.

V – Ventricular {V, E}

Ectopic focus in the ventricles: wide (≥ 120 ms), bizarre QRS, no preceding P wave, often an opposite-polarity T wave.

F – Fusion {F}

Normal and ventricular activation meet, giving an intermediate shape. Rarest class (< 1% of beats) and hardest to classify.

Labels are grouped into AAMI EC57 classes [12]; paced / unclassifiable beats (Q) are excluded, as in most MIT-BIH studies.

Literature Review & Research Gap

37 papers (2020–2026) on deep learning and XAI for ECG arrhythmia classification were reviewed.

Area	Existing direction & reported evidence
Deep learning models	CNN 96.8% vs CNN-LSTM 98.5% accuracy in a direct comparison [7]; lightweight CNN-LSTM with SHAP reports 98.24% over 9 rhythm classes [2]
Evaluation protocol	Only 30.3% of 122 studies use an inter-patient split [5]; where both are reported, F1 falls from 95.52% to 83.89% on unseen patients [4]
Explainability	Grad-CAM / SHAP mostly shown as example plots [2], [3]; < 10% of studies validate maps against P / QRS / T structures [6]

Research gap

Explanations are rarely quantified: do they fall on the right wave, are they faithful to the model, and do they stay stable when the ECG is noisy?

Our project position

Compare CNN vs CNN-LSTM + RR under intra- and inter-patient splits, then score Grad-CAM / SHAP with wave-region, deletion and noise-stability metrics.

Problem Statement & Motivation

Given a beat $x \in \mathbb{R}^{252}$ and its RR features r , learn a classifier $f(x, r) \rightarrow \hat{y} \in \{N, S, V, F\}$ together with an attribution map $A(x) \in \mathbb{R}^{252}$ that shows how much each sample drove the decision.

Black-box models

Thousands of learned weights and no stated reason; a clinician cannot verify the decision.

Class imbalance

$N \approx 90\%$, $S \approx 2\%$, $F < 1\%$ of beats: high accuracy can hide near-zero S / F recall.

Patient leakage

Beats of one patient look alike; random splits place them in both train and test sets.

Noisy signals

Muscle (EMG) and motion artefacts overlap the QRS band; most models are tested only on clean ECG.

Core pipeline: preprocess → classify → explain → validate

Motivation

- Long-term ECG screening needs automatic, beat-level arrhythmia detection.
- A heatmap a clinician can check (e.g. "wide QRS → V") builds trust in the prediction.
- Inter-patient and noise tests show how the model would behave on new, real recordings.

Project Objectives

Preprocess MIT-BIH

Band-pass filter, segment 252-sample beats around R peaks, compute normalised RR features, map labels to AAMI N / S / V / F.

Baseline & proposed

Train a 1D CNN baseline and a CNN-LSTM + RR model with class-weighted cross-entropy to handle the $\approx 45:1$ N:S imbalance.

Fair evaluation

Intra-patient (stratified 80/20) and inter-patient DS1 $\rightarrow$ DS2 splits; confusion matrix, per-class precision, recall, F1, macro-F1.

Explain & validate

Grad-CAM and Gradient SHAP maps; measure importance share on P / QRS / T and faithfulness with a deletion test.

Robustness & demo

Add NSTDB muscle noise at 18 / 6 / 0 dB SNR: F1 vs SNR and heatmap stability; Streamlit demo of prediction + heatmap.

Dataset: MIT-BIH Arrhythmia Database

MIT-BIH Arrhythmia Database (PhysioNet)

Moody & Mark [1] • 48 records × 30 min • 47 subjects • 2 leads • 360 Hz, 11-bit • ≈ 110,000 annotated beats

Data used

Lead MLII. Paced records 102, 104, 107, 217 removed
→ 44 records. Cardiologist beat labels and R-peak positions are used directly.

Two evaluation splits

Intra-patient: stratified random 80/20 split of beats.
Inter-patient [8]: train on DS1 (22 records), test on DS2 (22 other records).

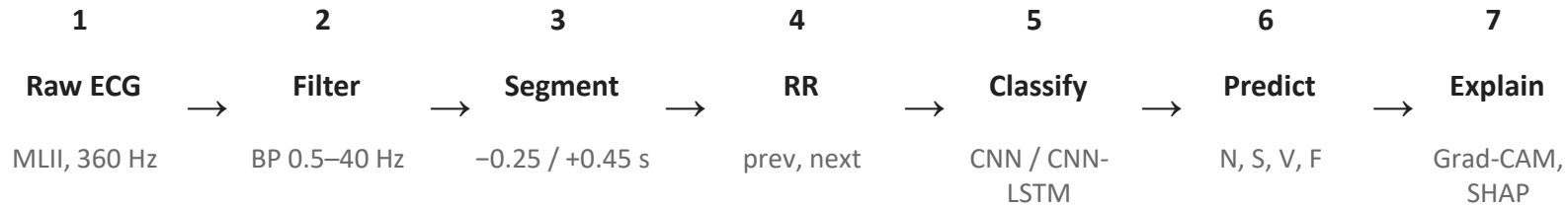
Split	N	S	V	F	Total
DS1 (train)	45,866	944	3,788	415	51,013
DS2 (test)	44,259	1,837	3,221	388	49,705

AAMI beat counts per split [8]: S and F together are < 3% of beats, so per-class F1 matters more than accuracy.

Noise data: MIT-BIH Noise Stress Test Database (NSTDB) [11]

The 'ma' (muscle artefact) recording is scaled and added to test beats at SNR = 18, 6 and 0 dB, where $SNR = 10 \cdot \log_{10}(P_{\text{signal}} / P_{\text{noise}})$.

Methodology



Steps 1–4 turn each record into fixed-length beats plus timing features; steps 5–7 classify each beat and produce a sample-level explanation.

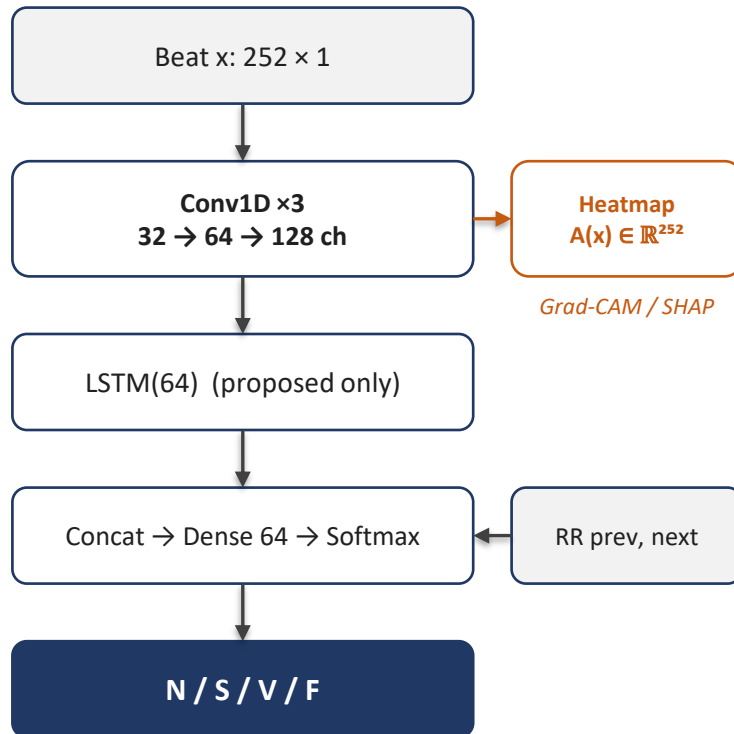
Preprocessing

- 4th-order zero-phase Butterworth band-pass, 0.5–40 Hz: removes baseline wander and 60 Hz mains
- Window: 90 samples before + 161 after the R peak = 252 samples (0.7 s)
- Per-beat z-score: $x' = (x - \mu) / \sigma$

Features & training

- RR_prev, RR_next divided by the record's mean RR (removes patient heart-rate differences)
- Class-weighted cross-entropy, $w_c = N / (K \cdot n_c)$
- Adam (lr 10^{-3}), batch 128, early stopping on validation macro-F1

Models & Explainability



Expected: V beats → heatmap on the wide QRS; S beats
→ more weight on the P wave and RR inputs.

Baseline: 1D CNN (≈ 44 k parameters)

- Conv1D 32 ($k=7$) → 64 ($k=5$) → 128 ($k=3$)
- Each: BatchNorm → ReLU → MaxPool(2): 252 → 31 steps
- Global avg. pool → Dense 64 → Softmax 4

Proposed: CNN-LSTM + RR

The 31×128 conv feature sequence feeds an LSTM(64) that models the P → QRS → T order; its last hidden state is concatenated with the 2 RR features before Dense 64 → Softmax 4.

Explainability (Captum)

- Grad-CAM [9]: $\alpha_k = (1/T) \sum_t \partial y_c / \partial A_k(t)$, $L = \text{ReLU}(\sum_k \alpha_k A_k)$, upsampled $31 \rightarrow 252$
- Gradient SHAP [10]: averages gradient $\times (x - x')$ over random baselines x' ; $\sum \phi_i \approx f(x) - E[f(x')]$

Evaluation & Explanation Metrics

Classification (per class)

Precision = $TP / (TP + FP)$, Recall = $TP / (TP + FN)$, $F1 = 2PR / (P + R)$. Macro-F1 = mean F1 over N, S, V, F, so rare classes count equally.

Region share (are maps clinical?)

$R_w = \sum_{t \in w} |A(t)| / \sum_t |A(t)|$ for $w \in \{P, QRS, T\}$.
Windows relative to R: P [-250, -80] ms, QRS [-50, +50] ms, T [+100, +400] ms.

Noise robustness

Add NSTDB muscle noise at $SNR \in \{18, 6, 0\}$ dB and plot macro-F1 vs SNR for both models.

Confusion matrix

4 × 4 table of true vs predicted class. Shows the typical errors: S beats called N (similar shape) and F beats called N or V.

Deletion test (are maps faithful?)

Zero the 10% highest-importance samples: $\Delta p = p_c(x) - p_c(x_{del})$. Compare with 10% random samples; a faithful map gives a much larger drop.

Explanation stability

ρ = Pearson correlation of $A(x_{clean})$ and $A(x_{noisy})$; $\rho \rightarrow 1$ means the explanation does not change with noise.

All metrics are reported for both the intra-patient and inter-patient (DS1 → DS2) splits.

Work Done Till Mid-Semester

Completed

- Literature review of 37 papers; research gap identified
- Framework, splits (incl. DS1 / DS2) and metrics defined
- V1: baseline 1D CNN, class-weighted loss, Grad-CAM
- V2: 5 step-by-step experiments + threshold tuning
- Final model: **98.41% accuracy, 91.10% macro-F1**

Development dataset [13]

Kaggle version of MIT-BIH, already segmented by the dataset authors: 187-sample beats (1.5 s at 125 Hz), scaled 0–1, five AAMI classes (N, S, V, F, Q).

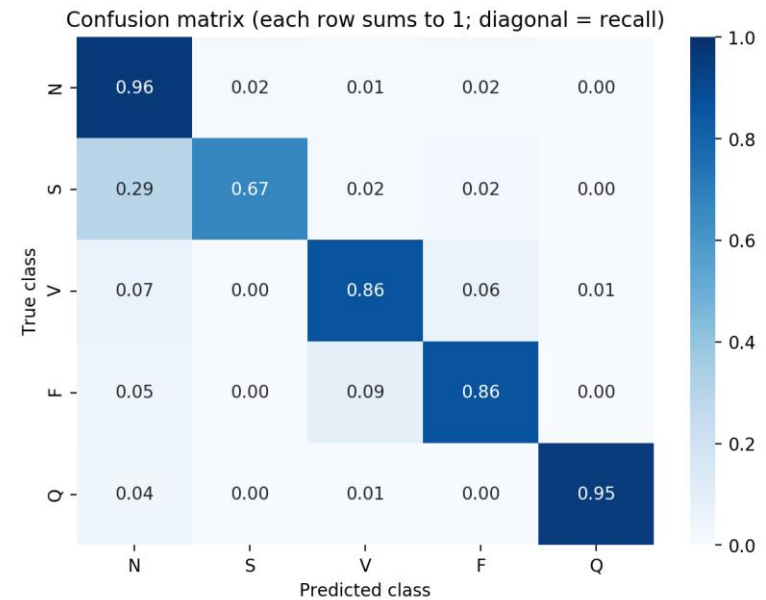
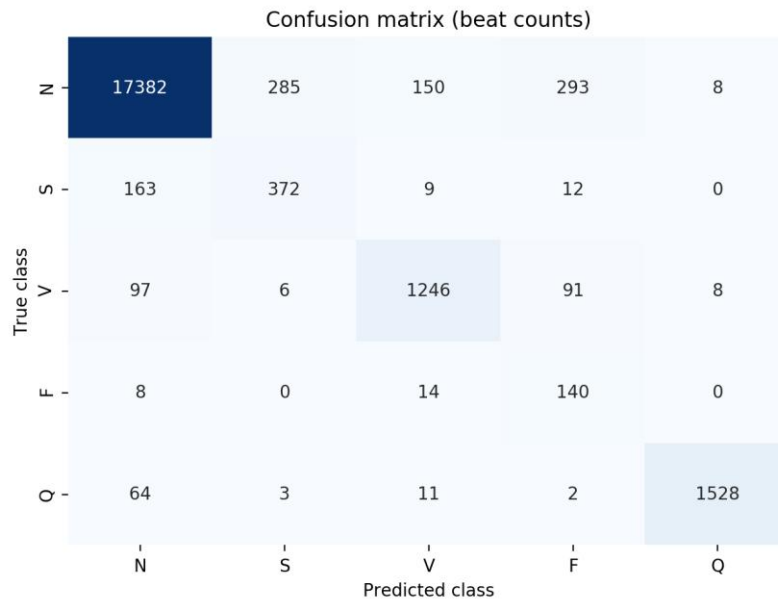
Used to build and debug the training, evaluation and Grad-CAM pipeline before our own MIT-BIH preprocessing.

File	N	S	V	F	Q	Total
Train file	72,471	2,223	5,788	641	6,431	87,554
Test file	18,118	556	1,448	162	1,608	21,892

Train file split 80/20 (stratified, seed 42) → 70,043 train / 17,511 validation beats; the test file is used only for final scores.

Caveat: the split is by beat, so the same patients appear in training and test data (intra-patient). These scores are optimistic; DS1 / DS2 comes next.

V1 Result: Baseline 1D CNN



Class	Precision	Recall	F1
N	0.981	0.959	0.970
S	0.559	0.669	0.609
V	0.871	0.861	0.866
F	0.260	0.864	0.400
Q	0.990	0.950	0.970
Macro	0.732	0.861	0.763

Accuracy 94.41% | Macro-F1 76.29%

- 538 beats predicted F, only 140 truly F (293 were N) → F precision 26%
- 29% of S beats called N (163 / 556)
- Validation accuracy swung 14.5% ↔ 95.1%; early stop at epoch 17 (best: 11)
- Cause: balanced weights make 1 F beat count as 113 N beats

V2: Step-by-Step Improvements

Same data, split and seed as V1. One change added per step; epoch, model and thresholds chosen on validation only.

ID	Change added	Acc. %	Macro-F1 %	S F1 %	F F1 %	Δ Macro-F1
B0	V1 baseline	94.41	76.29	60.90	40.00	–
E1	Best epoch by val. macro-F1	96.28	83.96	63.48	68.12	+7.67
E2	+ LR 5e-4, clipping, BN mom. 0.9	93.58	78.30	49.13	58.49	–5.66
E3	+ square-root class weights	97.39	87.58	69.84	76.97	+9.28
E4	+ light augmentation	97.02	86.20	66.16	76.25	–1.38
E5	+ residual CNN [14] (185 k params)	98.29	90.77	81.22	78.53	+4.57
Final	E5 + tuned thresholds (N × 3)	98.41	91.10	82.14	79.23	+0.33

What helped

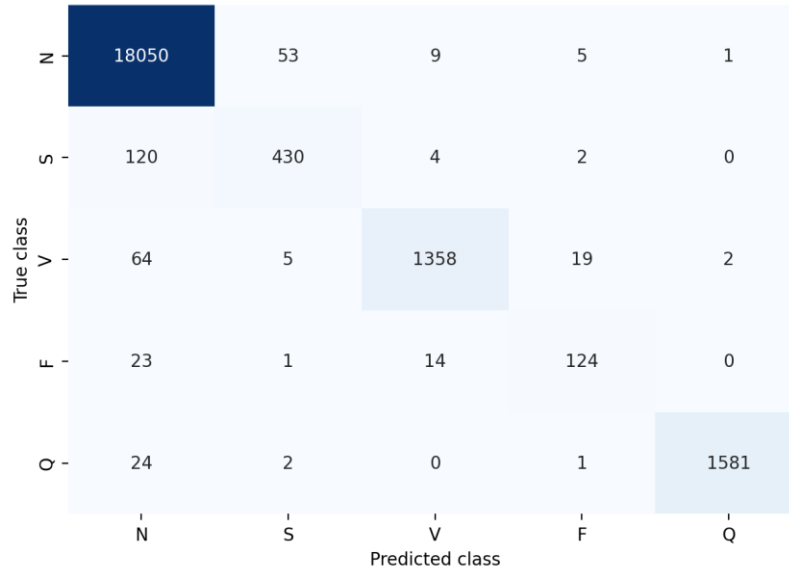
- E1: picking the epoch on macro-F1, not val. loss
- E3: softer weights (F:N 113:1 → 10.6:1) – biggest gain, F precision 44 → 76%
- E5: residual network – smoothest training, S F1 66 → 81

What did not

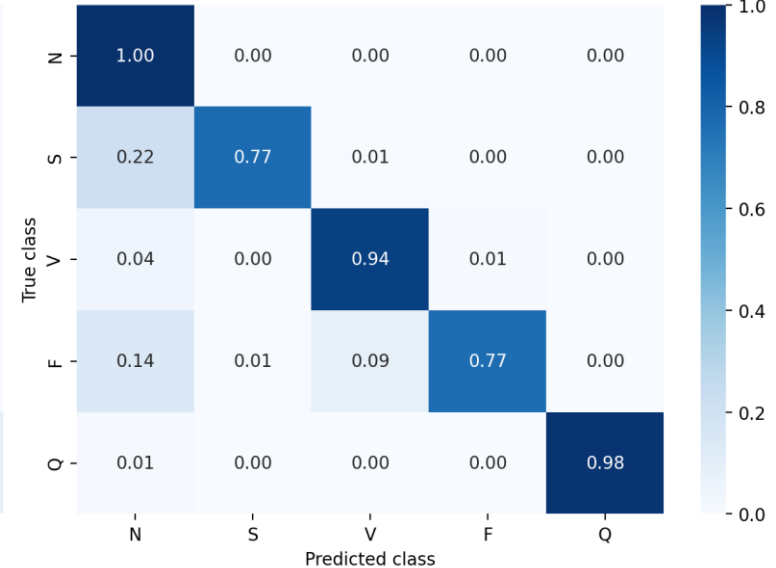
- E2 stability settings: –5.66, swings remained
- E4 augmentation: –1.38
- Threshold tuning: only +0.33, trades S / F recall for precision
- Steps are cumulative, so E2 / E4 were kept in the final model

Final Model: Residual CNN + Tuned Thresholds

Final model: confusion matrix (beat counts)



Final model: each row sums to 1; diagonal = recall



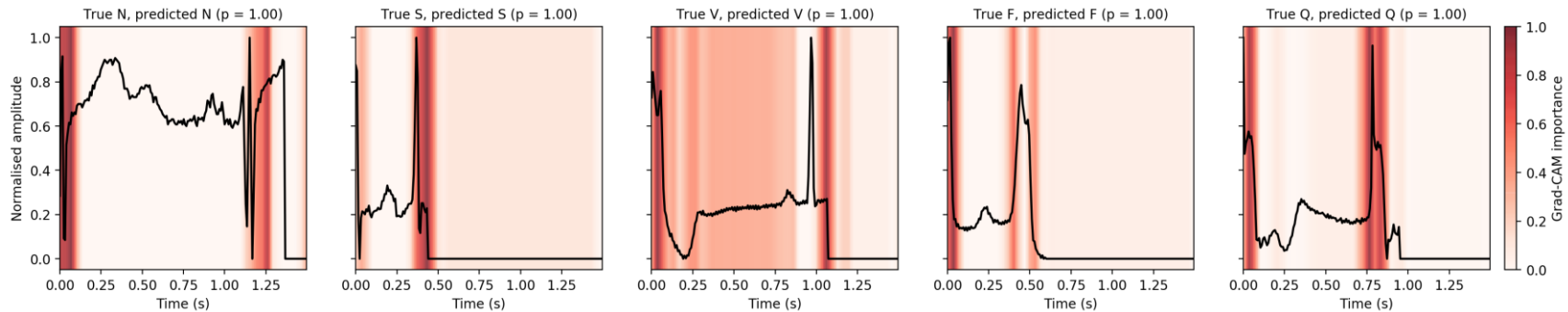
Class	Precision	Recall	F1
N	0.987	0.996	0.992
S	0.876	0.773	0.821
V	0.981	0.938	0.959
F	0.821	0.765	0.792
Q	0.998	0.983	0.991
Macro	0.933	0.891	0.911

Metric (%)	V1 (B0)	Final
Accuracy	94.41	98.41
Macro-F1	76.29	91.10
Precision S	55.86	87.58
Precision F	26.02	82.12
Recall S	66.91	77.34
Recall F	86.42	76.54

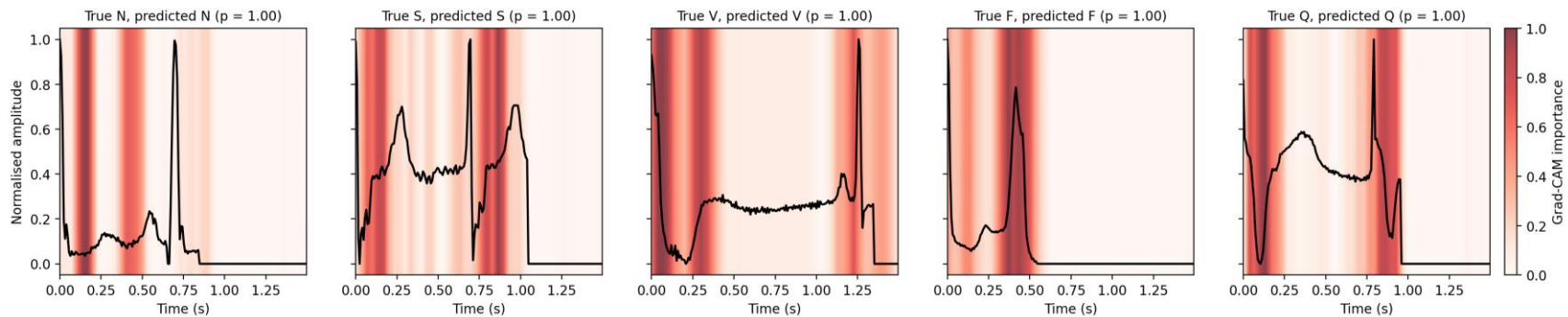
Largest remaining error: $S \rightarrow N$ (120 of 556 S beats). S differs from N mainly in timing, which motivates the RR features. F recall fell because V1's high recall came from over-predicting F.

Grad-CAM Explanations

V1 baseline CNN



Final model (residual CNN)



- Most confident correct test beat per class; red = regions that drove the prediction.
- Heat concentrates at the window start and around the second R peak → the model likely uses the timing of the next beat (an implicit RR cue), not only beat shape. Final-model maps are broader.
- Qualitative only so far → next: P / QRS / T region share and deletion test, plus SHAP.

Challenges & Lessons Learned

Class imbalance

Balanced weights (F:N 113:1) made the model over-predict F: precision 26%. Square-root weights (10.6:1) fixed most of it.

Unstable training

Validation accuracy swung 14.5% $\leftrightarrow$ 95.1%; early stopping on validation loss stopped too early. Selecting on macro-F1 and a residual CNN helped.

Not every change helps

Stability settings (E2) and augmentation (E4) lowered macro-F1. Steps were cumulative, so the best combination is not yet known.

Optimistic setting

Pre-segmented Kaggle beats, intra-patient split, 5 classes, single seed. Next: raw MIT-BIH, 4 classes, RR features and DS1 / DS2.

Conclusion And Future Scope

Key takeaways

- A working pipeline is in place: training, macro-F1-based evaluation and Grad-CAM.
- Macro-F1 76.29% → 91.10% (accuracy 94.41% → 98.41%) via epoch selection, softer class weights and a residual CNN.
- S → N remains the main error; results are intra-patient, so inter-patient scores will be lower.
- Published 98–99% accuracies are mostly intra-patient; our framework adds region, deletion and noise-stability scores.

Next steps

Raw MIT-BIH preprocessing (4 classes, RR features) → CNN vs CNN-LSTM + RR → DS1 / DS2 evaluation → Grad-CAM / SHAP with region and deletion checks → NSTDB noise tests → Streamlit demo.

Future scope

External test on INCART, automatic R-peak detection (Pan-Tompkins), MC-dropout uncertainty, full P/QRS/T delineation and clinician review of explanations.

References

- [1] G. B. Moody and R. G. Mark, "The impact of the MIT-BIH Arrhythmia Database," IEEE Eng. Med. Biol. Mag., vol. 20, no. 3, pp. 45–50, 2001.
- [2] N. Alamatsaz et al., "A lightweight hybrid CNN-LSTM explainable model for ECG-based arrhythmia detection," Biomed. Signal Process. Control, vol. 90, 2024.
- [3] D. Tenepalli and T. M. Navamani, "Design of an explainable deep learning framework for ECG-based arrhythmia prediction," IEEE Access, vol. 13, 2025.
- [4] Q. Xiao et al., "Deep learning-based ECG arrhythmia classification: A systematic review," Applied Sciences, vol. 13, 4964, 2023.
- [5] G. A. L. Silva et al., "A systematic review of ECG arrhythmia classification: Adherence to standards, fair evaluation, and embedded feasibility," arXiv:2503.07276, 2025.
- [6] N. N. Kulkarni et al., "Recent advances in deep learning based arrhythmia classification using ECG and PPG signals: a systematic review," Discover Computing, vol. 29, 2026.
- [7] P. Pramkeaw et al., "Deep learning-based ECG arrhythmia detection system: A comparative study of CNN, RNN, and LSTM architectures," IEEE ICCL, 2026.
- [8] P. de Chazal, M. O'Dwyer and R. B. Reilly, "Automatic classification of heartbeats using ECG morphology and heartbeat interval features," IEEE Trans. Biomed. Eng., vol. 51, no. 7, 2004.
- [9] R. R. Selvaraju et al., "Grad-CAM: Visual explanations from deep networks via gradient-based localization," Proc. IEEE ICCV, 2017.
- [10] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," Proc. NeurIPS, 2017.
- [11] G. B. Moody, W. Muldrow and R. G. Mark, "A noise stress test for arrhythmia detectors," Computers in Cardiology, vol. 11, 1984.
- [12] ANSI/AAMI EC57, "Testing and reporting performance results of cardiac rhythm and ST segment measurement algorithms," AAMI, 2012.
- [13] M. Kachuee, S. Fazeli and M. Sarrafzadeh, "ECG heartbeat classification: A deep transferable representation," Proc. IEEE ICHI, pp. 443–444, 2018.
- [14] K. He, X. Zhang, S. Ren and J. Sun, "Deep residual learning for image recognition," Proc. IEEE CVPR, pp. 770–778, 2016.